Policy Analysis - Randomization

March 3, 2026

Policy Evaluation - Estimating Treatment Effects

- With no independence:

$$ATE = \pi ATT + (1-\pi)ATU$$

- Do some algebra:

$$ATE_{est} = ATE + \underbrace{\{Avg_n[Y_i^0|D_i = 1] - Avg_n[Y_i^0|D_i = 0]\}}_{\text{Selection Bias}}$$

$$+ \quad \underbrace{(1-\pi)(ATT - ATU)}_{\text{Heterogeneous Treatment Effect Bias}}$$

Policy Evaluation - Estimating Treatment Effects

- Some notes on selection bias:

Policy Evaluation - Estimating Treatment Effects

- Some notes on selection bias:
  - Sometimes called *omitted variable bias*.

- Some notes on selection bias:
  - Sometimes called *omitted variable bias*.
  - Problem because selection into treatment depends on factors that are correlated with potential outcomes.

- Some notes on selection bias:
  - Sometimes called *omitted variable bias*.
  - Problem because selection into treatment depends on factors that are correlated with potential outcomes.
  - To the extent that we can observe and control for these correlates, then our estimates are free from bias (e.g., economy).

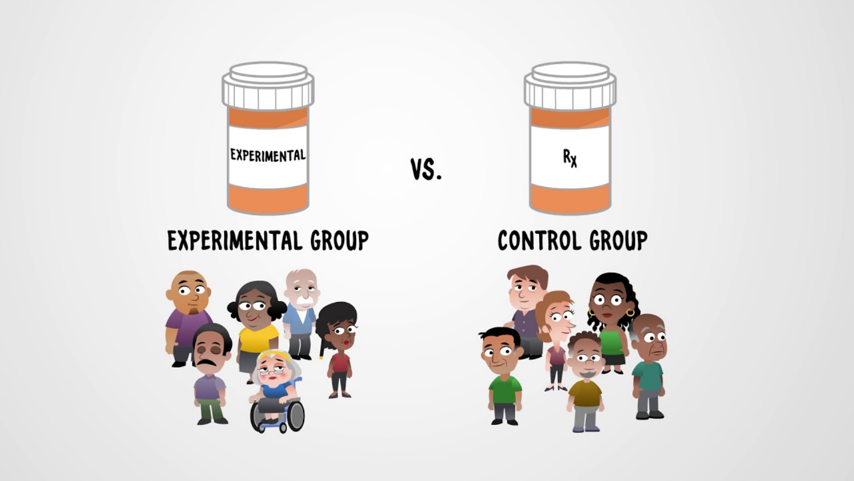
- Some notes on selection bias:
  - Sometimes called *omitted variable bias*.
  - Problem because selection into treatment depends on factors that are correlated with potential outcomes.
  - To the extent that we can observe and control for these correlates, then our estimates are free from bias (e.g., economy).
  - If these correlates are unobserved, then our estimates will be biased (e.g., risk aversion).

Policy Evaluation - Estimating Treatment Effects

- So how do we mitigate selection bias?

Policy Evaluation - Estimating Treatment Effects

- So how do we mitigate selection bias?



Policy Evaluation - Randomization

- Why does randomization mitigate omitted variable bias?

Policy Evaluation - Randomization

- Why does randomization mitigate omitted variable bias?
- If D is randomly assigned, then $(Y^1, Y^0) \perp\!\!\!\perp D$:
  - 1. $Avg_n[Y_i^0 | D_i = 1] = Avg_n[Y_i^0 | D_i = 0]$
  - 2. $ATT = ATU$

$$ATE_{est} = ATE + \underbrace{\{Avg_n[Y_i^0 | D_i = 1] - Avg_n[Y_i^0 | D_i = 0]\}}_{\text{Selection Bias} = 0} + \underbrace{(1-\pi)(ATT - ATU)}_{\text{Heterogeneous Treatment Effect Bias} = 0}$$

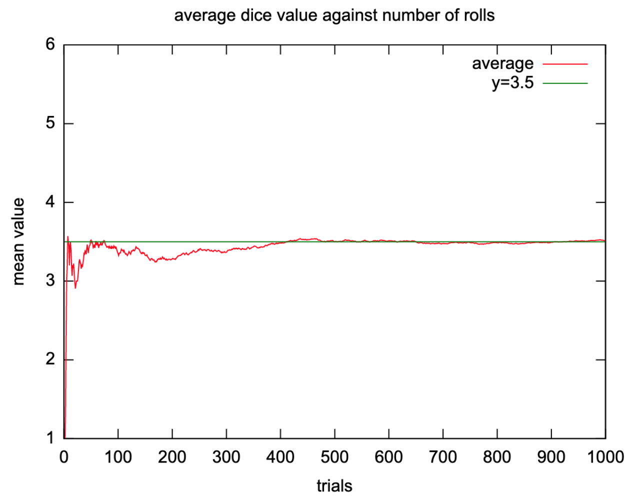
- Why does randomization mitigate omitted variable bias?
- If D is randomly assigned, then $(Y^1, Y^0) \perp\!\!\!\perp D$:
  - 1. $Avg_n[Y_i^0|D_i = 1] = Avg_n[Y_i^0|D_i = 0]$
  - 2. $ATT = ATU$

$$ATE_{est} = ATE + \underbrace{\{Avg_n[Y_i^0|D_i = 1] - Avg_n[Y_i^0|D_i = 0]\}}_{\text{Selection Bias} = 0}$$
$$+ \quad \underbrace{(1-\pi)(ATT - ATU)}_{\text{Heterogeneous Treatment Effect Bias} = 0}$$

- **IMPORTANT:** This only works when the sample is "sufficiently large".

Policy Evaluation - Randomization

- Law of Large Numbers:



Policy Evaluation - Randomization

- Randomization and the LLN are even more powerful than they seem.

Policy Evaluation - Randomization

- Randomization and the LLN are even more powerful than they seem.
- Sample Average Treatment Effect $=$
  $Avg_n[Y_i^1|D_i = 1] - Avg_n[Y_i^0|D_i = 0]$

Policy Evaluation - Randomization

- Randomization and the LLN are even more powerful than they seem.
- Sample Average Treatment Effect = $Avg_n[Y_i^1|D_i = 1] - Avg_n[Y_i^0|D_i = 0]$
- If n is "sufficiently large", then we can replace conditional *averages* with conditional *expectations*.

Policy Evaluation - Randomization

- Randomization and the LLN are even more powerful than they seem.
- Sample Average Treatment Effect = $Avg_n[Y_i^1|D_i = 1] - Avg_n[Y_i^0|D_i = 0]$
- If n is "sufficiently large", then we can replace conditional *averages* with conditional *expectations*.
- Population Average Causal Effect = $E[Y_i^1|D_i = 1] - E[Y_i^0|D_i = 0]$

Policy Evaluation - Randomization

- Randomization and the LLN are even more powerful than they seem.

- Sample Average Treatment Effect = $Avg_n[Y_i^1|D_i = 1] - Avg_n[Y_i^0|D_i = 0]$

- If n is "sufficiently large", then we can replace conditional *averages* with conditional *expectations*.

- Population Average Causal Effect = $E[Y_i^1|D_i = 1] - E[Y_i^0|D_i = 0]$

- Note that "sufficiently large" depends on the population mean and standard deviation of the outcome of interest.

Policy Evaluation - Randomization

- So randomization is the ideal and the "gold standard" of research methodologies.

Policy Evaluation - Randomization

- So randomization is the ideal and the "gold standard" of research methodologies.
- But...in policy analysis it's often (though not always) impossible to randomize individuals in our sample into treatment?

Policy Evaluation - Randomization

- So randomization is the ideal and the “gold standard” of research methodologies.
- But...in policy analysis it’s often (though not always) impossible to randomize individuals in our sample into treatment?
- How can we attempt to identify causal effects of policy when randomization is infeasible?